gebeya

Real-World Applications of Machine Learning

Objectives

- Explore how ML is used in real life
- Understand the impact of ML across industries
- Recognize everyday tools and services powered by ML

Why Real-World Applications Matter

- ML is embedded in our daily routines
- From your phone to hospitals to banks
- Understanding these apps help you connect the theoretical concepts with practical, measurable ways

ML in Entertainment – Netflix & YouTube

- The system tracks what you watched and for how long
- Learns your taste over time
- The more you watch, the more accurate

ML in E-Commerce – Amazon & Shopify

- **Product Recommendations**
  Recommends products you're likely to buy
- **Dynamic Pricing**
  Predicts demand and adjusts pricing
- **Customer Support**
  Powers chatbots and answers questions

ML in Finance – Fraud Detection

Behavioral Analysis
Checks if this matches your normal

spending behavior

Anomaly Detection
Flags the transaction

Real-time Alerts
Alerts you or the bank immediately

ML in Healthcare

Disease Prediction

Analyze patient data to predict certain conditions

Diagnostic Assistance

Assists doctors diagnose quickly and accurately

Treatment Personalization

Personalizes treatment plans based on what has worked

ML in Transportation – Google Maps & Uber

Traffic Prediction
Predicts traffic and travel times

Route Optimization
Suggests fastest routes

Arrival Estimation
Estimates when the ride will arrive and how long it will take

ML in Social Media – Instagram & TikTok

Content Curation

Decides which posts and videos you see first

Engagement Learning

Learns what content you engage with most

Network Expansion

Suggests accounts or friends you might know

ML in Agriculture

Yield Prediction
Predicts crop yields

Disease Detection
Detects plant diseases early

Planting Optimization
Suggests best planting times based on weather pattern

ML in Email – Gmail & Outlook

1

Spam Filtering

Filters out spam automatically

2

Smart Replies

Suggests quick replies

3

Behavior Learning

Learns what types of emails you open or ignore

ML in Voice Assistants – Alexa, Siri, Google Assistant

- Converts spoken words to text
- Learns your voice and preferences
- Improves with every interaction

ML in Education – Personalized Learning

Adaptive Platforms

Platforms adjust lessons based on student performance

Targeted Practice

provide extra practice problems or explanations

Intervention Prediction

Predicts which students need additional support

ML in Security – Facial Recognition

Identity Verification
Identifies individuals from images or video

Security Applications
Used in airports, phones, and public spaces

Adaptive Recognition
Trained to recognize faces under different lighting

ML in Manufacturing – Predictive Maintenance

Failure Prediction

Predicts when machines will break down

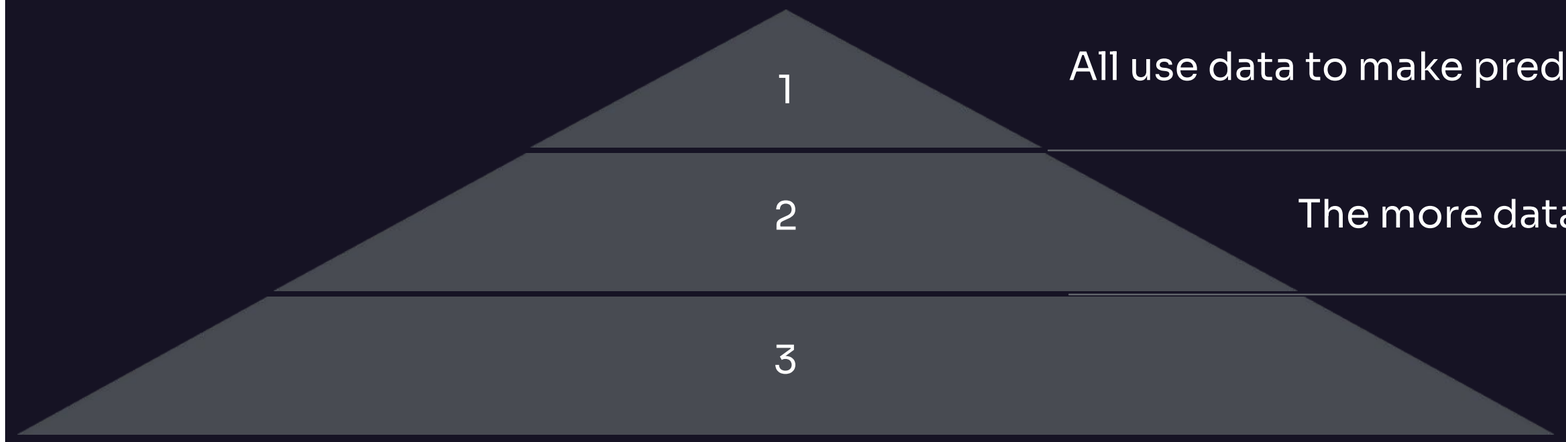
Cost Reduction

Fix issues before they cause costly downtime

Production Optimization

Keeps production running smoothly

The Pattern Behind All These



All use data to make predictions

The more data, the better the performance

The goal is faster, smarter, more personalized decisions

Summary

- ML powers many of the tools you use daily, like Netflix, Google Maps, and Gmail.
- It’s used in industries ranging from healthcare and finance to agriculture and manufacturing.
- ML improves personalization, speed, and accuracy in countless processes.
- It helps detect fraud, make medical diagnoses, and optimize logistics.
- Most applications rely on learning from user behavior or past data to improve over time.

Reflection Questions

1. Name three apps or platforms you use that rely on ML every day.
2. How does ML help in healthcare or finance?
3. What's one way ML improves your experience online?
4. Why does ML improve with more data?
5. Can you think of a new area where ML could be applied effectively?

Recommended Resources

1. Real- Life Examples of Machine Learning

https://www.geeksforgeeks.org/machine-learning/real-life-applications-of-machine-learning/

1. 9 Real-Life Machine Learning Examples

https://www.coursera.org/articles/machine-learning-examples

Thank you!

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